

CATALYST_GUIDE_v0.3.13

Version

v0.3.13-catalyst-workflow

Purpose

This version builds on earlier work and **formally locks in the catalyst workflow**. The aim is to allow isolated testing of specific, news-driven tickers (catalysts) such as STTK and LPSN without altering or breaking the normal top-gapper workflow.

- Implemented via `scripts/run\_catalyst.py` with a **safe universe swap**.
- Ensures that enrichment and backtests run as usual but on catalyst tickers only.
- After execution, the normal universe file is restored so no changes affect the main codebase.

Approach

1. **Catalyst runner** (`scripts/run\_catalyst.py`):

- Inputs: `--date`, `--scenario`, `--symbols` or `--file`.
- Writes a dated catalyst universe file (`data/catalyst/universe\_.txt`).
- Temporarily swaps this into `data/samples/universe\_sample.txt`.
- Runs enrichment (`fetch\_minutes\_polygon.py`) + backtest (`midas\_v2.cli` or `run\_day\_simple.py` fallback).
- Restores the original universe afterward.

2. **Design principles**:

- No edits to core code required.
- Catalyst output is stored in the normal `out/YYYYMMDD` structure.
- Compatible with existing summarizers and analysis tools.

What We Tested

STTK — 2025-08-05 (Scenario D)

Command:

...

```
python scripts/run_catalyst.py --date 2025-08-05 --scenario D --symbols STTK --fetch-minutes
```

...

- Enrichment wrote `data/samples/sample\_2025-08-05\_STTK.csv`.
- Backtest saved:
- `out/20250805/D/results\_2025-08-05.csv`
- `out/20250805/D/summary\_2025-08-05.txt`
- STTK result confirmed: **TP +2.04%**.

LPSN — 2025-08-06 (Scenario B)

Command:

...

```
python scripts/run_catalyst.py --date 2025-08-06 --scenario B --symbols LPSN --fetch-minutes
```

...

- Output saved to `out/20250806/B/results\_2025-08-06.csv` and summary.

Commands Used

- **STTK catalyst run**:

...

```
python scripts/run_catalyst.py --date 2025-08-05 --scenario D --symbols STTK --fetch-minutes
```

...

- **LPSN catalyst run**:

...

```
python scripts/run_catalyst.py --date 2025-08-06 --scenario B --symbols LPSN --fetch-minutes
```

...

- **Sanity check**:

...

```
$csv = 'out\20250805\D\results_2025-08-05.csv'; if (Test-Path $csv) { "$csv -> " + ((Get-Content $csv | Measure-Object -Line).Lines) }
```

...

Current Status

- Catalyst workflow verified.
- Safe swap + restore confirmed.
- Catalyst-specific enrichment and backtests validated with STTK and LPSN.
- Existing top-gapper code unaffected.

Next Stage

1. **Multi-day Catalyst Runs**

- Extend to multiple days (e.g., Aug 5 and Aug 6) and scenarios (B, D, E).
- Benchmark catalyst-only results against normal universe runs.

2. **Polygon Catalyst Integration**

- Enrich catalysts with Polygon news data (e.g., `\v2/reference/news``).
- Record headlines and catalyst type alongside each run.

- Use this enrichment to evaluate whether news-driven catalysts outperform random gappers.

3. **Filtering & Hygiene**

- Add blocklists (e.g., exclude Chinese tickers).
- Optional `--exclude-file` in `run_catalyst.py` for safe filtering.
- Paranoia mode (`--only-catalyst`) to ensure only chosen symbols are backtested.

4. **Docs & Summaries**

- Merge catalyst results into existing summary reports.
- Tag outputs clearly as catalyst runs for historical analysis.

Summary

v0.3.13-catalyst-workflow finalizes the safe universe swap design and confirms catalyst backtests with STTK and LPSN. The next focus is expanding catalyst validation across days/scenarios and enriching catalysts with Polygon news data, while keeping the main codebase stable.